

Venkateswara Reddy Pochana

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Experience

Dow Silicones Deutschland GmbH — *Master Thesis Researcher, Data Science* **May 2024 – Jul 2025**

- Led statistical analysis of polymer manufacturing data across 10+ process checkpoints to identify key quality drivers with over 70% correlation to product quality and optimize operating parameters through targeted experiments.
- Developed and validated machine learning models using historical polymer production data to predict final product properties, achieving 85% prediction accuracy and reducing quality drift by 50%.
- Owned experimental data standardization across EMEA research teams, establishing consistent data workflows and driving adoption through workshops and hands-on training, reducing manual data preparation by 20%.

Carl Zeiss AG — *Data Engineer Intern* **Dec 2023 – Apr 2024**

- Designed and deployed an automated document processing pipeline using Python, NLP-based entity extraction, Regex, and PDF processing for 10,000+ monthly invoices, reducing manual effort by 70%.
- Deployed Azure Data Factory and Databricks ETL pipelines to integrate multiple carrier and third-party logistics (3PL) APIs into Azure Cosmos DB, enabling near real-time shipment tracking and a unified view of logistics data.
- Built and owned Power BI dashboards to track logistics KPIs and shipment performance, helping reduce delivery delays by 20% and improve supply chain visibility.

Bosch Rexroth AG — *Business Intelligence Intern* **Mar 2023 – Aug 2023**

- Designed a master data management solution with historical tracking and built FastAPI APIs integrating Azure SQL and MongoDB data to ensure accurate billing for 2,000+ customers.
- Owned the end-to-end automation of invoice generation and reporting using Microsoft Power Platform (Power Automate, Power Apps, Power BI) and Python, reducing errors by 60% and saving 35 hours of manual work per month.

Amazon — *Project Management Intern* **Jul 2022 – Dec 2022**

- Built AWS-based ETL pipelines using S3, Lambda, and Redshift to automate ingestion and processing of EMEA food safety compliance audit data, centralizing 100K+ records for analytics and reporting.
- Led the development of AWS QuickSight dashboards for EMEA food safety compliance, unifying audit and ticketing data to improve KPI visibility for 100+ stakeholders and contributing to a 20% reduction in audit findings.

Divya Industries — *Production Engineer* **Apr 2019 – Oct 2020**

- Led data-driven process optimization, increasing manufacturing efficiency by 15% and reducing operating costs by 10%
- Integrated laser cladding into manufacturing processes, improving wear resistance and extending product life by 30%

Education

M.Sc. Scientific Instrumentation — Ernst-Abbe-Hochschule Jena, Germany	2025
B.Tech. Mechanical Engineering — JNTU Hyderabad, India	2019

Skills & Languages

Programming	Python, SQL
Data Engineering	ETL/ELT, Apache Spark, Databricks, Azure Data Factory, dbt, Apache Airflow, Apache Kafka
Cloud & Data Platforms	Microsoft Azure, AWS, Microsoft Fabric, Snowflake, Azure Synapse Analytics, Delta Lake
Databases	PostgreSQL, MySQL, MongoDB, Azure Cosmos DB
Machine Learning & AI	Scikit-learn, TensorFlow, PyTorch, Generative AI, RAG, LLMs
BI & Analytics	Power BI, Tableau, AWS QuickSight
Development & Automation	FastAPI, Git, Docker, Microsoft Power Platform (Power Automate, Power Apps, Power Pages)
Languages	English (C1), German (B2)